

Supply Chain Performance & Analytics Dashboard

Prepared By

Shahadat Hossain Sajim

Data Analyst

PostgreSQL • Supabase • Power BI • Dax • GitHub

Version 1.0.0 | September 2026

Executive Summary

This project analyzes end-to-end supply chain performance across procurement, inventory, production, and sales using SQL and Power BI.

The objective was to bring together operational data from different supply chain areas and turn it into a clear, interactive dashboard that highlights business performance, operational risks, product profitability, inventory conditions, and production quality.

The analysis was performed using SQL on data hosted in Supabase/PostgreSQL, followed by the development of a four-page Power BI dashboard for business-focused reporting.

Key Business Metrics

Metric	Result
Total Revenue	\$176.95M
Total Profit	\$48.56M
Total Units Sold	187K+
Total Production	3.83M
Profit Margin	27.44%
Inventory Risk Rate	29.17%
Production Defect Rate	0.63%

These metrics provide a high-level view of overall sales performance, profitability, production output, inventory risk, and production quality.

Key Findings

Sales & Product Performance

- **Galaxy S24 Ultra** was the strongest product in terms of both revenue and total profit.
- **Galaxy Z Fold5** was another major contributor to overall revenue and profitability.
- Several home appliance and TV products showed strong profit margins, indicating that high profitability was not limited to smartphone products.
- Product-level sales volume analysis helped identify products with the highest demand.

Procurement & Supplier Performance

- Procurement analysis highlighted differences in supplier spending, procurement volume, cost, and lead time.
- Supplier-level analysis helps identify areas where procurement spending and lead-time performance should be monitored more closely.
- Product-level procurement analysis provides visibility into which products require the highest procurement quantities and costs.

Inventory Performance

- 29.17% of current inventory records were identified as being at or below their reorder points, indicating potential replenishment risks.
- Inventory risk varied significantly across facilities.
- Mumbai DC showed the highest inventory risk, followed by New Jersey DC, while Poland DC had a comparatively lower risk level.
- Current inventory and estimated inventory value analysis helps prioritize products that require closer inventory monitoring.

Production Performance

- Thai Nguyen Plant recorded the highest production output among the analyzed facilities.
- Production quality analysis showed an overall defect rate of approximately 0.63%.
- Galaxy S24 Ultra had the highest number of defective units, followed by products such as Galaxy Buds2 Pro and Galaxy Watch6 Classic.
- Defective-unit analysis provides a more actionable view of production quality than relying only on overall defect rates.

Business Recommendations

Based on the analysis, the following areas should receive continued attention:

1. Monitor inventory risk at facilities with a higher proportion of stock at or below reorder points.
2. Review high-defect products to identify potential production or quality-control issues.
3. Prioritize high-revenue and high-profit products when planning inventory and production availability.
4. Review supplier cost and lead-time performance to identify opportunities for procurement improvement.
5. Use the dashboard regularly to monitor changes in sales, inventory, production, and procurement performance.

Dashboard Overview

The final Power BI dashboard is organized into four pages:

Page 1 — Executive Overview

Provides a high-level view of revenue, units sold, profit, production output, revenue trends, top products, facility production, and product profitability.

Page 2 — Procurement & Supplier

Focuses on procurement spending, procurement quantity, procurement cost, supplier performance, and supplier lead time.

Page 3 — Inventory & Production Performance

Highlights inventory risk, current inventory levels, production output, defective units, and estimated inventory value.

Page 4 — Sales & Product Performance

Analyzes product profitability, product-line performance, sales volume, and revenue by product category.

Analytical Approach

The project followed a structured analytics workflow:

Data Collection → Supabase/PostgreSQL → Data Validation → SQL Analysis → Data Modeling → DAX Measures → Power BI Dashboard → Business Insights

SQL was used to answer the core business questions and validate the underlying data, while Power BI was used to present the results through interactive dashboards and KPIs.

Conclusion

The analysis provides a consolidated view of supply chain performance across procurement, inventory, production, and sales.

The dashboard makes it easier to identify sales and profitability drivers, inventory risks, supplier performance differences, and production quality issues from a single reporting environment.

Overall, the project demonstrates how SQL and Power BI can be used together to transform operational supply chain data into practical, business-focused insights for decision-making.